Akash Patil

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PROFESSIONAL SUMMARY

DevOps & Site Reliability Engineer with 2+ years of experience designing, deploying, and maintaining scalable **AWS** & **Azure** based infrastructure. Skilled in **CI/CD automation**, **infrastructure as code(Terraform & Ansible)**, and containerized workloads using **Jenkins**, **Docker**, and **Kubernetes**. Experienced in **monitoring, centralized logging, alerting**, and **incident response** using **Splunk, Prometheus, Grafana, Amazon CloudWatch** and **Azure Monitor**, ensuring reliable, secure, and highly available production systems.

TECHNICAL SKILLS

Languages: C, C++, JavaScript, Python, SQL

Technologies/Frameworks: HTML5, CSS3, React.js, Express.js, Node.js, MongoDB

Devops Tools/Technologies: Git, GitHub, GitHub Actions, Linux, Shell Scripting, Networking, AWS, Azure, Docker, Kubernetes, Ansible, Terraform, Jenkins, CI/CD, Splunk, Prometheus, Grafana, SDLC

Cloud Platforms:
AWS (EC2, S3, RDS, EKS, ECR, CloudWatch, Lambda, CodePipeline, API Gateway, Bedrock)
Azure (AKS, ACR, Azure DevOps, Azure Monitor, Azure Functions, App Insights, Key Vault, AI Foundry, APIM, Logic Apps)

EXPERIENCE

Equations Work | Software Engineer | Pune, India **Mar 2024 - Present**

Cloud Continuum Platform (CCP) | AWS CodePipeline, Terraform, Amazon Bedrock, API Gateway

* Architected an end-to-end **CI/CD platform** on **AWS CodePipeline** managing infrastructure, AI agents, and frontend delivery from a single source of truth, reducing deployment efforts by **60%** across dev, QA, and production environments
* Provisioned complete **AWS infrastructure** using **Terraform** via CodePipeline, covering **Amazon Bedrock**, **API Gateway**, AWS Amplify, CloudWatch, **Lambda**, and **Step Functions** with remote state stored in **S3**, eliminating all manual infrastructure setup
* Built and automated **Python**-based **AI agent** lifecycle - packaging, versioning, and deploying intelligent agents into **Amazon Bedrock** via CodePipeline, ensuring consistent environment-specific delivery with **automated rollback** on failure
* Secured enterprise API access by registering Bedrock endpoints in **Amazon API Gateway** via Terraform, enforcing **IAM** authentication, rate limiting, and request logging, reducing unauthorized access incidents by **40%**
* Implemented **AI quality governance** using **AWS Lambda** and **Step Functions**, auto-creating **ServiceNow** tickets on failures and reducing **MTTR by 35%**
* Established full-stack observability using **Amazon CloudWatch** across Bedrock, API Gateway, Amplify, and Lambda, improving incident detection time by **30%**

Agentic Yodha | Genworks Health | AWS EKS, ArgoCD, Helm, Gemini AI, CloudWatch

* Designed and deployed a **cloud-native AI platform** on **Amazon EKS** comparing customer documents using **FastAPI**, **Gemini AI**, and **agentic workflows**, reducing document review time by **70%**
* Implemented end-to-end **CI/CD pipelines** using **AWS CodePipeline** and **CodeBuild** to automate build, test, and containerization workflows, increasing deployment frequency by **3x** across environments
* Implemented **GitOps**-based continuous delivery using **ArgoCD** and **Helm** charts for environment-specific deployments, achieving **zero-downtime** rolling updates and automated rollback on failure
* Secured application secrets using **AWS Secrets Manager** with **Kubernetes** secret injection, eliminating hardcoded credentials entirely across all environments
* Enabled **Horizontal Pod Autoscaling (HPA)** on **EKS** to handle variable AI processing workloads, maintaining **99.9% uptime** and reducing infrastructure costs by **25%**

* Established end-to-end observability using **Splunk, Amazon CloudWatch, AWS X-Ray**, and **Container Insights** for centralized log analysis, proactive alerting, and faster root-cause analysis, reducing **MTTR by 35%**

DevSecOps CI/CD Pipeline | Johndeere | Terraform, Jenkins, Azure AKS, Argo CD, Prometheus

* Implemented an end-to-end **DevSecOps CI/CD pipeline** for a **containerized MERN stack application** deployed on **Azure Kubernetes Service (AKS)**
* Automated cloud infrastructure provisioning (**VNet**, **AKS cluster**, node pools, **Azure RBAC**, **Application Gateway** integration) using **Terraform**, orchestrated via **Jenkins**
* Built a secure **CI pipeline** integrating **SonarQube** quality gates, **OWASP** dependency checks, and **Trivy** container vulnerability scanning
* Containerized frontend, backend and database services and managed image lifecycle using **Docker** and **Azure Container Registry (ACR)**
* Implemented **GitOps**-based continuous delivery with **Argo CD**, deploying **Kubernetes manifests** with automated rollouts
* Enabled production-grade observability using **Splunk, Prometheus**, and **Grafana** by centralizing application and Kubernetes logs, accelerating production troubleshooting.

EDUCATION

PVG's College Of Engineering and Technology (Affiliated to SPPU) **CGPA: 9.09**
Bachelor Of Engineering in Information Technology *July 2020 - July 2024*

PROJECTS

Azure Infrastructure Automation with Terraform | Terraform, Jenkins, Azure, Python Flask, MySQL

· Automated **CI/CD** via **Jenkins** to deploy a Flask REST API with **Azure Database for MySQL**, ensuring seamless updates

· Provisioned scalable **Azure infrastructure** (VNet, Azure VMs, Azure Load Balancer) using **Terraform**, enforcing security via subnet isolation and **Azure RBAC**

Cloud Resource Tracker | Shell Scripting, AWS CLI, Cron, JQ

· Automated tracking of **AWS resources (EC2, S3, Lambda)** via Shell scripts and AWS CLI, using JQ for data parsing

· Implemented **scheduled Cron jobs** for consistent reporting and **cost optimization**

Repo-Access Automator | Shell Scripting, GitHub API, Curl

· Developed a **shell script** using **GitHub API** and **Curl** to **automate** repository access management, with secure API integration

· Enhanced script with **input validation** and **modular structure**, ensuring robustness for production use